

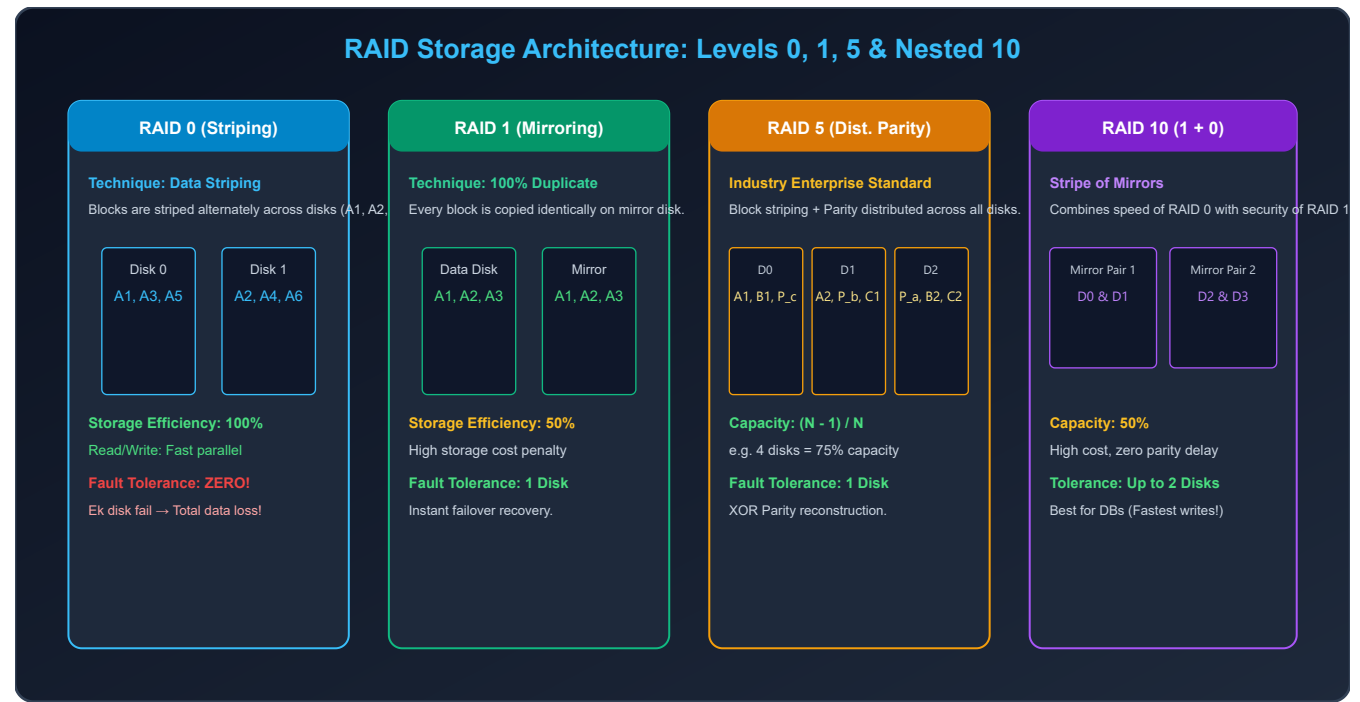
Module 06: RAID Architecture — Levels 0 to 6

1. RAID Core Techniques

Striping: Parallel I/O speed • **Mirroring:** 100% duplicate copy • **Parity:** XOR mathematical recovery ($D_1 \oplus D_2 = P$).

2. RAID Levels Comparison Table

RAID Level	Technique	Usable Capacity	Fault Tolerance	Target Workload
RAID 0	Block Striping	$N \times \text{DiskSize}$ (100%)	0 (Zero)	High throughput, non-critical
RAID 1	Mirroring	50% Capacity	1 disk per pair	OS boot disks, accounting
RAID 5	Distributed Parity	$(N - 1) \times \text{DiskSize}$	1 disk failure	Standard enterprise servers
RAID 6	Dual Parity (P+Q)	$(N - 2) \times \text{DiskSize}$	2 simultaneous failures	Critical cloud storage
RAID 10	Stripe of Mirrors	50% Capacity	Up to 2 disks	High-performance Databases



3. Key Viva Question

Why is RAID 10 preferred over RAID 01?

RAID 10 me disk failure par sirf single mirror recover hota hai; do alag pairs se simultaneous disk failures survive ho sakte hain.